Activity 1 AII501NAA
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1. Spam Filter Email Agent (50%)

a)
Task Environment:

Performance Measures:
- +1 when email directory contains emails from allow list
- +1 when restrict list &/ bad word list remove spam
- -1 when spam is detected in email directory
- +1 when spam directory only contains spam from restrict,bad word list.
- -1 when email detected in spam directory
- -1 when spam detected in allow list

Environment:
- Allow list
- Bad word list
- Restrict list
- Spam Directory
- Email Directory

Actuator:
- Access allow list, bad word list, restrict list
- Scan lists
- Verify emails
- Scan email & spam directory
- Send spam to spam directory if verified as true

Sensors:
- Type “text” readable by the agent
- .eml format w/ header and body
- Header plain text
- Body encoded/encrypted text

b)

Types:	Answer:
Observable??	Sensors can perceive current states of the environment

Deterministic??	Deterministic; Directory environment state changes
Episodic??	Sequential; Follows one action set, receives new data, doesn't derive from past experience
Static??	Dynamic; Agent changes the directory environment; an external factor(updated directory) changes the environment
Discrete??	Continuous; email files can be broad in terms of content
Single-agent??	Multiple agents can maximize performance

c)
A simple reflex agent works because the task at hand doesn't require a specific end goal. The task is for the agent to filter the email directory for spam continuously. The main sensors would perceive the world based on the state, evolution, and the agent's actions in that world, then determine an action afterwards to perform in the environment.

d) Check repo

2. Search Problem (50%)

a)

States:
variables x,y,z are integers
State = (x,y,z)
x = (0<=x<=12) = jug1
y = (0<=y<=8) = jug2
z = (0<=z<=3) = jug3

Initial:
Empty
x=0 (jug1)
y=0 (jug2)
z=0 (jug3)
(0,0,0)

Goal State:
x=1 (jug1)

y=1 (jug2)
z=1 (jug3)

Transition model & Actions:
Fill from faucet into jug of choice:

AddFaucet(x, p)
Legal if:
Jug 1(x) p>=1, 0<p+x<=12

Jug 2(y) p>=1, 0<p+y<=8

Jug 3(z) p>=1, 0<p+z<=3

Pour into jug of choice:
PourInto(x,y, p)
Legal if:
Jug 1(x) p<= x, 0<p+y<=8
Jug 1(x) p<= x, 0<p+z<=3

Jug 2(y) p<= y, 0<p+x<=12
Jug 2(y) p<= y, 0<p+z<=3

Jug 3(z) p<= z, 0<p+x<=12
Jug 3(z) p<= z, 0<p+y<=8

Empty onto ground:
EmptyGround(x, p)
Legal if:
Jug 1(x) 0<p<=x
Jug 2(y) 0<p<=y
Jug 3(z) 0<p<=z

Results Logic:
p variable is applied to jugs through pouring by subtracting water from the source jug, then applying the water it subtracted to the next jug.
For instance:
pourJug12(x(the variable of jug1)-p, p+jug2(state of jug2)).

b)
A heuristic can work due to its ability to search for the right path under a lot of simulated actions. The problem requires us to determine a single outcome from 3 sources through 3 primary

actions. Heuristic is helpful here due to it being able to find the distance to the target value, in our case, the gallon target size between the 3 jugs.

c)
No, not all search algorithms can work. The problem wants us to determine how to get 1 gallon from any one of the 3 jugs. A search algorithm can work because it would involve determining the least number of actions to find the best path to the goal. Breadth works as an example due to being able to explore all possibilities, then finding the shortest path to the destination. However, algorithms that can't show repeated states aren't reliable in some cases. For instance, DFS has an issue of getting stuck looping in itself in deep cycles without proper checks. Therefore, not all search algorithms are good for this problem.

d) Check repo